MEGHANA RAJAKUMAR

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Creative Technologist · Enterprise AI Solutions · Partner-Facing Technical Lead · XR for Manufacturing & Training

Technical Project Lead and hands-on engineer with 6+ years designing, building, and delivering AI, XR, and cloud solutions at global enterprises including BMW Group and Epsilon (Publicis Groupe). At BMW, I take enterprise use cases from stakeholder discovery and prototyping through pilot and production, leading delivery while building the software myself. My work spans vision and generative AI, from fine-tuning NVIDIA's NVILA vision-language model for automated quality inspection to RAG with semantic search for BMW's agentic commerce app in ChatGPT; full-stack development in Python/FastAPI, Angular, and React; and XR development in Unity (C#), with Meta Quest 3/Pro builds optimized to 90 FPS. As liaison between business stakeholders, engineering, design, and external partners, I translate workflows and pain points into requirements and working software, and run the demos that move tools into production. Earlier at Epsilon, I led AWS and Azure cloud migrations and built Terraform, Ansible, and CI/CD automation.

CORE COMPETENCIES

Partner & Solution Leadership
- Partner Enablement & Technical Documentation
- Agile deliveries
- Solution Design & Architecture
- Reviews
- Stakeholder & Partner Management
- SDK & Developer Tooling Development
- Prototyping to Production Scaling
- Product Roadmap & Strategy

AI, Physical AI & Cloud
- SLM & LLM Fine-Tuning for Physical AI
- Vision-Based AI Workflows
- Edge & On-Device AI Deployment
- Agentic AI Workflows
- Dataset Curation & Domain-Specific Model Training
- Cloud Automation (AWS, Azure), CI/CD & DevOps

XR & Wearable
- Enterprise XR Solution Architecture
- AR for Manufacturing & Training
- Smart Glasses & Wearable XR
- Unreal Engine & Unity (C#, Plugin Architecture)
- Digital Twin Development & Simulation Design
- Real-Time 3D Pipelines: Modeling, Texturing, Rigging, Animation
- XR Performance Optimization

Platforms & Tools: Unity, Unreal Engine, Meta Quest 3/Pro, Meta Horizon OS (Android APK pipeline), C++, C#, AWS, Azure, Terraform, Ansible, gRPC, FastAPI, and TypeScript/React

PROFESSIONAL EXPERIENCE

IT Technical Project Lead and Technical Artist

BMW Group – Greenville, South Carolina **Nov 2023 - Present**

- **Designed and built SmartEye Video Processor end to end as sole developer**, a self-service data preparation platform that centralizes video dataset collection, preprocessing, and standardization for AI model training across BMW developer teams. Developed the Python/FastAPI backend and Angular frontend using Density (BMW's design system) to segment uploaded video into dataset groups, transcribe and label content, and apply configurable transforms and anonymization to protect the identities of people on camera. Owned client demos, user support, handover, and industrialization.
- **Built 3D content and interactions for Impossible Battery, a digital-twin XR training simulation of BMW's fifth-generation high-voltage EV battery for Meta Quest 3/Pro**, now live across BMW plants worldwide to train technicians on assembly and disassembly of a high-risk component. Modeled, textured, rigged, and animated 70+ assets and programmed their C# interactions in Unity. Demoed to internal teams and external partners including Meta; the project was featured by Meta and Unity.
- **Owned Meta Quest 3/Pro testing and performance optimization**, profiling each build on both devices to catch bugs and interaction issues, and reducing APK size while holding 90 FPS to prevent motion sickness. Joined client feedback sessions after each test build and turned requested changes into the next iteration.

- **Led discovery and built the frontend for an AI-driven Legal Defense platform** (5-person team) that consolidates hundreds of case documents into a single vehicle timeline and flags discrepancies and evidentiary gaps. Interviewed technicians and field analysts to map BMW's litigation review workflow as process SME, and built the prototype that demonstrated time savings on a real case for Gateway review. Developed the React frontend from UX designs, integrated backend APIs, coordinated design and backend teams, and ran beta testing. Now in pilot, with a conservative projected **ROI of $22M/yr.**
- **Fine-tuned and evaluated vision models for SmartEye Human Amplification,** BMW's Vision AI system that automates a 160+ step vehicle quality checklist previously done manually over days. Compared YOLO and vision-language models for video action detection, then labeled production-line footage and fine-tuned NVIDIA's NVILA to recognize technician actions, auto-check completed steps, and flag missed ones. Tested egocentric wearables (Meta Ray-Ban, Seeed Studio) against fixed-position cameras for data capture and assessed edge feasibility for real-time inference, enabling hands-free QA while reducing latency and inference cost by 1.3x.
- **Contributed platform research, frontend optimization, and a capability framework to Customer Companion, BMW's agentic commerce app in ChatGPT**, now live in Europe. Assessed app-building capabilities across AI chat platforms including Microsoft Copilot and Google Gemini. Applied traditional RAG, semantic search embeddings, and progressive profiling to maximize personalization across customer evaluation and purchase decisions.
- **Acted as the primary technical point of contact for a formal external partnership with SCADPro**, guiding an academic partner team through the full integration lifecycle of a live BMW XR program – from early concept and architecture review through prototype testing to production rollout – translating partner feedback into technical requirements and running iterative demonstrations on weekly feedback cycles.

Software Engineer II and Cloud Automation Expert

Epsilon (Publicis Groupe) – Bangalore, India **Jul 2019 - Sept 2022**

- Designed, implemented, and scaled infrastructure automation and deployment pipelines using Terraform, Ansible, and CI/CD frameworks, driving a 20% reduction in deployment times and a 32% improvement in overall deployment efficiency by eliminating manual processes.
- Led complex cloud and production operations, including specialized migrations across AWS and Azure, end-to-end orchestration of critical workflows and data transitions, and major production release upgrades, identifying risks, defining mitigation strategies, and minimizing downtime while improving system stability and reliability.
- Owned production reliability and team enablement, rapidly resolving high-impact issues across production tenants to ensure high availability and consistent performance, while also supporting hiring efforts by interviewing entry-level technical candidates.

EDUCATION

Masters of Arts (3.7/4.0 GPA) **09/2022 – 06/2024**

Savannah College of Art & Design - Savannah, USA

- Major in Animation, Interactive Technology, Special Effects and Video Graphics

Bachelor of Engineering (4.0/4.0 GPA) **08/2015 – 07/2019**

RV College of Engineering - Bangalore, India

- Major in Computers and Telecommunication Engineering

PROFESSIONAL & ACADEMIC ACHIEVEMENTS AND AWARDS

- **Gold Medalist in Engineering**, awarded for outstanding academic excellence and top performance in the program
- **Interview Panelist** at the SCAD Faculty Conference, representing both BMW Group and SCAD; featured in the official SCAD Events Blog
- Published researcher with **411+ IEEE citations**, bridging applied research and real-world technology commercialization